Econ 605 Notes

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General

- GDP = flow. "bricks per quarter"
- Y_t = (stock/flow) total output - not always GDP
- C_t = (stock/flow) consumption
- K_t = (stock/flow) capital - sometimes includes IP, sometimes just physical
- X_t = (stock/flow) QUANTITY of capital...
- $m = e^{-\bar{r}(s,t)}$: discount factor - sometimes with integral????

Solow

- $\dot{A}_t$: (flow) wealth??

OLG

- ??

Ramsey

Planner's Problem

- H: total number of households
- l_t: household size (exogenous)
- $L_t = 1 * l_t * H$ (1 = labor supply per person)
- $C_t = Hx_t$
- x_t: household consumption - CONTROL! (state is K) but we replace this with:
- $c_t = \frac{C_t}{e^{gt} L_t}$
- $k_t = \frac{K_t}{e^{gt} L_t}$
- ρ: time discounting (impatience)
- θ: IES intertemporal elasticity of substitution
- $\theta = -\frac{u_c(c)}{c \cdot u_{cc}(c)} = -\frac{d \ln c}{d \ln(u'(c))}$
- $g_Y = n + g$
- $\frac{\dot{c}}{c} + g$: growth rate of per-capita consumption
- $\vec{\alpha} = (\rho, \theta, n, d, g)$ ALL ARE EXOGENOUS
- $PD(\vec{\alpha})$: phase diagram for comparative dynamics

From Optimal Control Slides:

- $\bar{k}$: max debt constraint or min wealth constraint IN TERMINAL STATE
- Γ: law of motion for capital/wealth/the state variable: $\Gamma(k_t, c_t, t) = \dot{k}$
- λ_t: marginal benefit of one unit of k_t
- $\lambda_t k_t$ = agent's wealth measured in units of utility (i.e., potential to receive future utility)

LOM for capital:

$$\dot{K} = I - dK = Y - C - dK$$

LOM for capital, intensive form:

$$\dot{k_t} = y_t - c_t - (n+d)k_t = \underbrace{f(k_t) - c_t}_{\substack{\text{actual investment} \\ \text{per effective worker}}} - \underbrace{(n+d+g)k_t}_{\substack{\text{replacement investment} \\ \text{per effective worker}}}$$

Production Function/Total Output

$$Y = F(K, e^{gt}L) = C_t + I_t$$

Planner's Problem

$$\underbrace{V(k_0, t)}_{\text{indirect utility}} = \max_{c_t} \int_t^\infty e^{\rho t} e^{nt} u(c_t) dt \text{ s to}$$

1. $\dot{k} = y_t - c_t - (n+d)k_t$
2. $k_0 > 0$ given
3. $\lim_{t \to \infty} \lambda_t k_t$ (transversality condition with exponential discounting)
4. implicitly: $k_t, c_t \geq 0$ - will never bind due to Inada so we usually ignore them

Planner's Problem FOCs:

- $\frac{\partial \mathcal{H}}{\partial c_t} = 0$
- $\frac{\partial \mathcal{H}}{\partial k_t} = -\dot{\lambda}$
- $\frac{\partial \mathcal{H}}{\partial \lambda_t} = \dot{k_t}$

Euler Equation for Consumption (Solution to the Ramsey Problem)

$$\frac{\dot{c}_t}{c_t} = \frac{1}{\theta}(r_t - (\rho + \theta g))$$

Phase Diagram Equations (steady state)

$$\dot{k} = 0 \iff c_t = f(k_t) - (n + d + g)k_t$$
$$\dot{c} = 0 \iff k_t = k_*, \text{ where } f'(k_*) = d + \rho + \theta g$$
$$c_* = f(k_*) - (n + d + g)k_*$$

Golden Rule vs Steady State Capital

- $f'(k_*) = d + \rho + \theta g$
- $f'(k_G) = d + n + g$

So

$$f'(k_*) > f'(k_G) \iff \rho > n + (1 - \theta)g$$

Isoclines

- $\dot{k} = 0$: $c_t = f(k_t) - (n + d + g)k_t$
- $\dot{c} = 0$: $r = \rho + \theta g$ or $k_t = k_*$, where $f'(k_*) = d + \rho + \theta g$

  Usually, $g = 0$, especially if you get the EE from the consumer max problem.

  λ never jumps at t_A (unless capital is destroyed which is rare) c always jumps at t_A (i.e., when there is a surprise). We don't know which way it jumps without more clues though.

Jump or no jump?

	$t = t_A$	$t = t_C$
Surprise change	N/A	c_t, λ_t typically jump
Pre-announced change	c_t, λ_t typically jump	c_t, λ_t typically continuous*

*Pre-announced change makes c_t, λ_t jump at the time of change if:

- Both c_t, λ_t jump if k_t jumps at time t_C for an exogenous reason (e.g. pre-announced confiscation of wealth)

- c_t jumps if parameter changes make u_c, Γ_c discontinuous (e.g. pre-announced jump in sales tax rate or a parameter change making marginal utility jump)

Algorithm to solve Ramsey Shocks

1. Transition K to k
2. Solve Planner's Problem
    1. Derive EE
    2. Check that you have two DEs (should look like this):
        1. $\frac{\dot{c_t}}{c_t} = \alpha k^{\alpha-1} - d - \rho$
        2. $\dot{k_t} = k_t^\alpha - c_t - (n+d)k_t$
3. Derive isoclines. How do they move (if at all)?
    1. $\dot{c} = 0 \Rightarrow k = (\frac{\alpha}{d+\rho})^{\frac{1}{1-\alpha}}$ (hint: make sure exponent is positive)
    2. $\dot{k} = 0 \Rightarrow c_t = k_t^\alpha - (n+d)k_t$
4. Is there a jump in c_t at t_c?
    1. Capital not destroyed, so co-state is continuous
    2. Set $\lambda_{t-0} = \lambda_{t+0}$ to see if c_t is continuous or not.
5. Draw diagram
    1. old/new curves + old/new arrows
    2. where should you be at t_{c-0} to be able the new stable arm?
    3. what is plausible behavior at $t = t_A$? Options are
        1. stay
        2. jump up
        3. jump down (can't always choose which one–sometimes either 2 or 3 is correct in drawings)
    4. Label jumps and their timing

Ramsey - Competitive Equilibrium/Household Expectations

- c_{it}: per-capita consumption (not consumption per effective worker)
- g: tech growth. We usually assume $g = 0$ - no technology growth in the consumer problem so the Euler equations of the Household and Planner coincide.
- ε_u: elasticity of Marginal Utility growth with respect to consumption growth. If ε_u rises, then consumption growth is less sensitive to changes in the interest rate. In other words, consumers prefer smooth consumption despite time-varying interest rates.
- $\Phi(K)$: an expectation/belief function that allows households to forecast future factor prices

Household Utility Maximization

$$\max_{c_{it}} \int_0^\infty e^{-\rho t} L_{it} u(c_{it}) dt \text{ s. to:}$$

1. $\dot{A_{it}} = r_t A_{it} + w_t L_t - c_{it} L_{it}$ (aka household flow budget constraint) DON'T USE $\dot{K}$ here!!
2. A_{i0}: given
3. $\lim_{T\to\infty} e^{-\bar{r}(0,T)} A_{it} \geq 0$

- Individual State Variable: A_{it}
- Aggregate state variables: can use r_t, w_t or K_t, L_t. Both will give the same answer, but we use K_t, L_t in this class.

Household's Utility Maximization Problem FOCs:

- $\frac{\partial \mathcal{H}}{\partial c_{it}} = 0 \iff$
- $\frac{\partial \mathcal{H}}{\partial A_{it}} = -\dot{\mu_{it}}$

Then eliminate μ_t, do some nasty rearranging and define $\varepsilon_u = |\frac{u''(c_{it})c_{it}}{u'(c_{it})}|$

Euler Equation for Consumption (Solution to the Ramsey Problem)

$$\frac{\dot{c_{it}}}{c_{it}} = \frac{1}{\varepsilon_u}(r_t - \rho) = -\frac{u'(c_t)}{u''(c_t)}(r_t - \rho)$$

This is identical to the Euler equation in the planner's problem if $g = 0$ and $\theta = \varepsilon_u$.

Ramsey Asset Market Clearing

$$A_t = K_t$$

Technically also have the capital leasing sector which requires

$$A_D = K_t \text{ and } L_D = L_t$$

Ramsey Capital leasing max problem Idk if this is important, so don't get sidetracked, but here it is:

$$\max_{A_D}(R_t - r_t - d)A_D$$

where A_D is the decision of the capital leasing firms.

Ramsey Goods Market Clearing

$$Y_t = C_t + I_t$$

Note: Asset Market Clearing implies goods market clearing in Ramsey

Def "Competitive (Walrasian) Equilibrium"

Competitive equilibrium is a list of quantities $\{C_t, Y_t, I_t, K_t, A_t\}$ and prices $\{w_t, r_t, R_t\}$ such that

1. $\{C_t, A_t\}$ solves the household problem given $\{w_t, r_t\}$
2. Production side equilibrium holds for all t
    1. $w_t = \frac{\partial Y_t}{\partial L_t}, R_t = \frac{\partial Y_t}{\partial K_t}, r_t = R_t - d$
3. Market clearing for all t
    1. $A_t = K_t \Rightarrow Y_t = C_t + I_t$

Def "Production Side Equilibirum"

A list of quantities $\{C_t, Y_t, K_t\}$ and prices $\{w_t, r_t, R_t\}$ satisfying, for all t:

1. Aggregate production function: $Y_t = F(K_t, L_t)$
2. Profit maximization and factor market clearing
    1. Final goods sector: $(K, L) \in argmax_{(K,L)}(F(K, L) - R_t K - w_t L)$ given (w_t, R_t)
    2. Value maximization and asset market clearing, capital leasing: $K_t \in argmax_K(\int_t^\infty e^{-\bar{r}(t,s)-d(s-t)} R_s K ds - K)$ given (r_t, R_t)
3. Free entry (automatic given profit maximization and market clearing)
    1. $\max_{(K,L)}(F(K, L) - R_t K - w_t L) = 0$

Def "Household Agrees with the Planner"

When we have (1) production side equilibrium, and (2) Asset market clearing ($A = K$), then planner and household have the same (1) utility, (2) LOM for wealth ($\dot{K}$), (3) transversality condition.

Careful: if we don't have asset market clearing and/or production side equilibrium, HH would have different resources and the stable arms would not coincide.

Def "Rational Expectations Equilibrium (REE)"

Refinement of WE - we add an expectation/belief function $\Phi(K) = c_t, C_t = \Phi(K_t)L_t$ that allows households to forecast future factor prices

Definition of equilibrium: REE is a list of quantities, $\{C_t, Y_t, I_t, K_t, A_t\}$, prices $\{w_t, r_t, R_t\}$ and $\Phi(K)$ satisfying

1. Utility maximization (*different from WE*) The consumption decision rule $c_i(a, K_t; \Phi(K_t))$ solves the household problem given the information households have at date t and their beliefs about behavior of others, $c_t = \Phi(K_t)$

Details: Households form factor price forecasts (function-to-function mappings) based of date-t state K_t and their beliefs Φ:

$$\{w^e_{t+\tau}\} = \mathcal{W}(K_t; \Phi(K_t)), \qquad \{r^e_{t+\tau}\} = \mathcal{R}(K_t; \Phi(K_t)), \text{all } \tau > 0.$$

Utility maximization problem:

$$c_i(a, K_t; \Phi) \in \underset{\{c_{it}\}}{\operatorname{argmax}} \int_0^\infty e^{-\rho t} L_{it} u(c_{it}) dt$$

$$\int_t^\infty \exp[-\bar{r}^e(t,s)] \cdot c_{is} \cdot L_{is} ds \leq a \cdot L_{it} + \int_t^\infty \exp[-\bar{r}^e(t,s)] \cdot w^e_s \cdot L_{is} ds$$

Definition of equilibrium – continued

2. Production-side equilibrium (no change)

3. Rational expectations (consistency) <u>conditions</u> <u>on Φ</u> (new)

Households' own consumption decision (c_i) based on *equilibrium* beliefs is consistent with their beliefs about others' consumption (Φ) for any initial condition $K_0 > 0$ and all $t \geq 0$:

$$c_i\left(\frac{K_t}{L_t}, K_t; \Phi(K_t)\right) = \Phi(K_t)$$

where K_t solves

$$\dot{K}_t = F(K_t, L_t) - \Phi(K_t)L_t - dK_t \text{ given } K_0 > 0, L_t = L_0 e^{nt}.$$

4. Asset market clearing (no change), follows from RE condition 3

FINISH MEE

Def "Representative Household flow budget constraint"

$$\dot{A}_t = r_t A_t + w_t L_t - C_t$$

Government and fiscal policy in the Ramsey model

Endogenous Growth

- A: (stock?) patent number (index)
- $P_{A,t}$: price of idea
- M (?) market value of patents
- $\dot{M}$: Capital gain
- φ: Knowledge externality. Stepping on toes or synergy.
- λ: Labor externality...
- θ: ????????????????????????????
- flow = quantity measured as bricks per quarter
- stock = total number of bricks

Misc

Def "Euler equation"

An Euler equation is something that relates choice in one period to choice in another period.

Remark "discrete to continuous, stocks vs flows"

REMEMBER HOW TO CONVERT - one of them gets multiplied by delta t

In steady state $\dot{K} \neq 0$!!!! ONLY RATIOS ARE UNCHANGING IN STEADY STATES!!

Prereqs

Def "Annual growth rate"

Def "Annualized growth rate"

Def "Instantaneous growth rate"

Fact "Arithmetic with growth rates"

- $g_{XY} = g_X + g_Y$
- $g_{X/Y} = g_X - g_Y$
- $g_{X^\alpha} = \alpha g_X$